

Role of mitochondrial oxidative stress to explain the different longevity between genders: protective effect of estrogens.



Females live longer than males. Work from our laboratory has shown that this may be due to the up-regulation of longevity-associated genes by estrogens. Estrogens bind to the estrogen receptors and subsequently activate the mitogen activated protein kinase and nuclear factor kappa B signalling pathways, resulting in an up-regulation of antioxidant enzymes. Estrogen administration, however, has serious undesirable effects and of course, cannot be administered to males because of its powerful feminizing effects. Thus, we tested the effect of genistein, a phytoestrogen of high nutritional importance whose structure is similar to estradiol, on the regulation of the expression of antioxidant, longevity-related genes and consequently on oxidant levels in mammary gland tumour cells in culture. Phytoestrogens mimic the protective effect of oestradiol using the same signalling pathway. The critical importance of up-regulating antioxidant genes, by hormonal and dietary manipulations, to increase longevity is discussed.